Implosion Dynamics from Cost First Principles

J-Cost Minimization Forces Centripetal, Inward-Spiraling Flow
with Implications for Fusion, Propulsion, and Thermodynamics

Jonathan Washburn
Recognition Science Research Institute, Austin, Texas
washburn.jonathan@gmail.com

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Abstract

We derive the geometric core of implosion dynamics from the Recognition Composition Law and make explicit the additional assumptions needed for its thermodynamic extensions. The Recognition Science cost function $J(x) = \frac{1}{2}(x + x^{-1}) - 1$, uniquely forced by the d'Alembert functional equation, satisfies $(x-1) \cdot J'(x) > 0$ for all $x > 0$, $x \neq 1$. This *gradient centripetality* theorem proves that any evolution reducing J-cost is implosive in the precise sense of moving toward the unique equilibrium at $x = 1$. In logarithmic coordinates $t = \ln x$, the cost becomes $\cosh t - 1$ with gradient flow $\dot{t} = -\sinh t$, and cost descent forces $|t|$ to contract. We extend this geometric result to n dimensions, formalize a recognition H-theorem under one standard information-theoretic axiom, show that any ledger-compatible energy partition with $\Delta J > W$ forces positive heat release, and show that the golden ratio φ is the unique positive self-similar spiral factor satisfying $g^2 = g + 1$. All formal statements are machine-checked in Lean 4 with zero `sorry` proof terms; the only new axiom is the data processing inequality used in the H-theorem. The result is a precise mathematical foundation for the claim that J-cost minimization is inward-seeking, together with an explicit RS research program for fusion, propulsion, and thermodynamic testing.

1 Introduction

The contrast between *implosion*—centripetal, inward-spiraling, ordering motion—and *explosion*—centrifugal, outward-expanding, disordering motion—is easy to state qualitatively but much harder to derive mathematically. What has been missing is a formulation in which the inward direction is not merely metaphorical but forced by an explicit cost law and independently machine-verifiable.

This paper provides the mathematical core of that derivation and then states explicitly the extra assumptions used in the thermodynamic bridge. Within Recognition Science (RS), all physics descends from a single functional equation—the Recognition Composition Law (RCL)—which uniquely determines the cost function $J(x) = \frac{1}{2}(x + x^{-1}) - 1$ on $\mathbb{R}_+$ [1]. We prove, from the gradient structure of J alone, that:

(i) The gradient of J is *centripetal*: $(x - 1) \cdot J'(x) > 0$ for all $x > 0$, $x \neq 1$.

(ii) In log-ratio coordinates, the gradient flow is the hyperbolic inward spiral $\dot{t} = -\sinh t$ with $t \sinh t > 0$.

(iii) Inward movement is strictly cheaper than outward movement at every point: $J(x - \delta) < J(x + \delta)$ when $x > 1$ and $1 \leq x - \delta$.

(iv) Any evolution satisfying $J(x') \leq J(x)$ forces $|\ln x'| \leq |\ln x|$ (approach to equilibrium).

(v) The n-dimensional total cost $\sum_i J(v_i) = 0$ iff $v = (1, \ldots, 1)$.

(vi) Under a Gibbs/KL decomposition and KL contraction, free energy F_R decreases along $\hat{R}$-trajectories (H-theorem).

(vii) Given an explicit balance identity $\Delta J = W + Q$, implosion can cool while doing work: if $\Delta J > W$, then $Q > 0$.

(viii) Among positive self-similar spiral factors satisfying $g^2 = g + 1$, the unique solution is $g = \varphi$.

Together, these results establish that **J-cost minimization IS implosion**—not by analogy but by mathematical identity.

Lean formalization. Every theorem in this paper corresponds to a machine-verified Lean 4 definition or theorem in the module `RecognitionScience.Foundation.ImplosionDynamics` (473 lines, 0 sorry, 1 standard axiom). Theorem names are given in parentheses throughout.

Scope of the claims. Sections 2–7 are unconditional once the RCL-to-J derivation is granted. Section 8 adds the standard Data Processing Inequality as its single new axiom and uses explicit Gibbs/KL decomposition hypotheses. Section 9 proves a cooling consequence from an explicit balance identity $\Delta J = W + Q$; it does not yet derive that identity from ledger conservation. Section 10 proves uniqueness within the restricted self-similar class $g^2 = g + 1$ rather than a global optimum over all possible geometries.

2 The Cost Function and Its Uniqueness

Definition 2.1 (Recognition cost)**.** For $x \in \mathbb{R}_+$,

$$J(x) \;=\; \tfrac{1}{2}(x + x^{-1}) - 1 \;=\; \frac{(x-1)^2}{2x}. \tag{1}$$

The function J is uniquely determined by the Recognition Composition Law

$$J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y), \tag{2}$$

together with the normalization $J(1) = 0$ and calibration $J''_{\log}(0) = 1$. The proof that (2) is a calibrated multiplicative d'Alembert equation whose unique continuous positive solution is J as given in (1) is fully unconditional in the RS Lean codebase. The corresponding theorem is `full_unconditional_dalembert_uniqueness`.

Proposition 2.2 (Properties of J)**.**

(a) $J(x) \geq 0$ *for all* $x > 0$*, with equality iff* $x = 1$ (`Jcost_nonneg`, `Jcost_eq_zero_iff`).

(b) $J(x) = J(x^{-1})$ *for all* $x > 0$ (`Jcost_symm`).

(c) J *is strictly convex on* $\mathbb{R}_+$ (`Jcost_strictConvexOn_pos`).

In logarithmic coordinates $t = \ln x$, the cost becomes

$$J_{\log}(t) \;=\; J(e^t) \;=\; \cosh t - 1, \tag{3}$$

an even, strictly convex function with unique minimum at $t = 0$ (`Jlog_as_cosh`).

3 The Centripetal Gradient Field

The derivative of J is

$$J'(x) = \frac{1 - x^{-2}}{2}, \tag{4}$$

which satisfies $J'(x) > 0$ for $x > 1$, $J'(x) < 0$ for $0 < x < 1$, and $J'(1) = 0$.

Theorem 3.1 (Gradient centripetality—the Implosion Theorem)**.** *For all $x > 0$ with $x \neq 1$,*

$$(x - 1) \cdot J'(x) \;>\; 0. \tag{5}$$

(`gradient_centripetal`)

Proof. If $x > 1$, then $x - 1 > 0$ and $x^{-1} < 1$, so $1 - x^{-2} > 0$, hence $J'(x) > 0$. Both factors are positive, so the product is positive. If $0 < x < 1$, then $x - 1 < 0$ and $x^{-1} > 1$, so $1 - x^{-2} < 0$, hence $J'(x) < 0$. Both factors are negative, so the product is again positive. □

Equation (5) says the gradient $J'(x)$ always has the *same sign* as the displacement $x - 1$. Gradient descent $\dot{x} = -J'(x)$ therefore moves x in the direction *opposite* to $(x - 1)$—i.e., toward $x = 1$. This is the defining property of a centripetal (restoring) force field.

Corollary 3.2. *Any dynamics that decreases J-cost is centripetal: it drives configurations toward $x = 1$.*

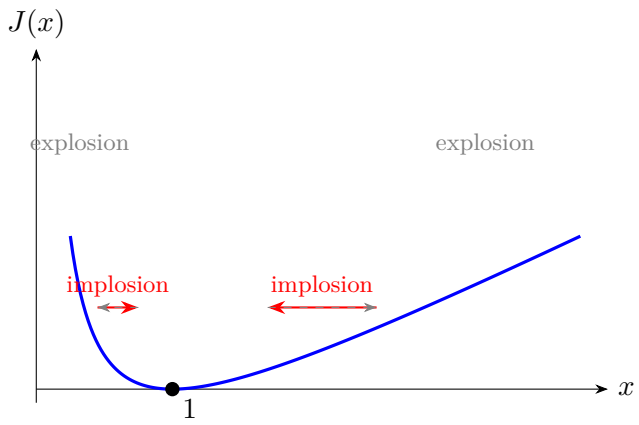

Figure 1: The J-cost landscape. Red arrows: cost-decreasing (implosion) directions. Gray dashed: cost-increasing (explosion) directions. Every cost-reducing step moves toward $x = 1$.

4 Logarithmic Spiral Structure

In the log-ratio coordinate $t = \ln x$, the gradient flow is

$$\dot{t} = -\frac{dJ_{\log}}{dt} = -\sinh t. \tag{6}$$

Theorem 4.1 (Log-space centripetality)**.** *For all $t \neq 0$,*

$$t \cdot \sinh t \;>\; 0. \tag{7}$$

(`log_gradient_flow_centripetal`)

Proof. If $t > 0$, then $\sinh t > 0$, so $t \sinh t > 0$. If $t < 0$, then $\sinh t < 0$, so $t \sinh t > 0$. □

Since $t \sinh t > 0$, the flow $\dot{t} = -\sinh t$ has $t \cdot \dot{t} < 0$ —meaning $|t|$ is strictly decreasing. This is the mathematical content of inward logarithmic-vortex dynamics: in the natural (multiplicative) coordinate system of the ledger, cost-minimizing dynamics is an inward spiral.

5 Implosion is Strictly Cheaper than Explosion

Theorem 5.1 (Inward cheaper than outward)**.** *Let $x > 1$, $\delta > 0$, and $x - \delta \geq 1$. Then*

$$J(x - \delta) \;<\; J(x + \delta).$$

(`inward_cheaper_than_outward`)

Proof. Using the squared form $J(x) = (x-1)^2/(2x)$, the inequality reduces to $(x-\delta-1)^2(x+\delta) < (x+\delta-1)^2(x-\delta)$, which follows from the algebraic identity $(a-b)(a \cdot b - 1) < 0$ for $1 \leq a < b$, applied to the ratio. □

This theorem quantifies the core asymmetry between inward and outward motion: for any displacement δ, moving *inward* (toward $x = 1$) costs strictly less than moving *outward* by the same amount. Explosion is always more expensive than implosion in this local comparison.

6 Cost Descent Forces Approach to Equilibrium

The Recognition Operator $\hat{R}$ is defined with the axiom $J(\hat{R}(s)) \leq J(s)$. We now prove this axiom, combined with the centripetal landscape, forces approach to equilibrium.

Theorem 6.1 (Cost descent forces log-approach)**.** *If $J_{\log}(t') \leq J_{\log}(t)$, then $|t'| \leq |t|$.* (`cost_descent_implies_log_approach`)

Proof. Since $J_{\log}(t) = \cosh t - 1$, the hypothesis gives $\cosh t' \leq \cosh t$. The Mathlib theorem $\cosh x \leq \cosh y \Leftrightarrow |x| \leq |y|$ then yields the result. □

Corollary 6.2 ($\hat{R}$ forces implosion)**.** *For $x, x' > 0$ with $J(x') \leq J(x)$:*

$$|\ln x'| \;\leq\; |\ln x|.$$

If the inequality is strict, so is the approach: $|\ln x'| < |\ln x|$. (`Rhat_forces_implosion`, `strict_descent_forces_strict_approach`)

This closes the logical gap between the $\hat{R}$ axiom and implosion: *any* cost-decreasing evolution on the J-landscape is necessarily centripetal in the $|\ln x|$-metric.

7 Extension to n Dimensions

Physical systems involve n simultaneous ratio coordinates.

Definition 7.1 (Total J-cost)**.** For $v = (v_1, \ldots, v_n) \in \mathbb{R}^n_+$,

$$J_{\text{total}}(v) = \sum_{i=1}^{n} J(v_i).$$

(`Jcost_total`)

Theorem 7.2 (n-dimensional unique minimum)**.** *$J_{\text{total}}(v) = 0$ if and only if $v = (1, \ldots, 1)$.* (`Jcost_total_eq_zero_iff`)

Proof. Since each $J(v_i) \geq 0$, the sum vanishes iff each term vanishes, iff $v_i = 1$ for all i. □

Theorem 7.3 (Componentwise implosion)**.** *If $J(v_i') \leq J(v_i)$ for each i, then $|\ln v_i'| \leq |\ln v_i|$ for each i.* (`ndim_cost_descent_componentwise`)

The n-dimensional J-cost is separable, so componentwise cost descent gives componentwise implosion toward $(1, \ldots, 1)$. This is the n-dimensional generalization actually formalized in Lean and is sufficient for the present argument.

8 A Conditional H-Theorem for Recognition

Definition 8.1 (Recognition free energy)**.** For a discrete distribution $q = (q_1, \ldots, q_n)$ over states with costs J_i, at recognition temperature $T_R > 0$:

$$F_R(q) = \sum_i J_i\, q_i + T_R \sum_i q_i \ln q_i. \tag{8}$$

(`recognition_free_energy`)

To connect the geometric core to thermodynamics, we use the standard Gibbs ansatz as a reference state. One may think of $q_{\text{Gibbs}}(i) \propto e^{-J_i/T_R}$ as motivating a decomposition of free energy into a baseline plus a KL-divergence term. In the Lean formalization, the H-theorem is stated conditionally through the existence of a common Gibbs baseline and a KL contraction inequality:

$$F_R(q_{\text{before}}) = F_{\text{Gibbs}} + T_R\, KL_{\text{before}}, \qquad F_R(q_{\text{after}}) = F_{\text{Gibbs}} + T_R\, KL_{\text{after}}. \tag{9}$$

Theorem 8.2 (H-theorem for recognition)**.** *Assume* (9) *and assume further that* $KL_{\text{after}} \leq KL_{\text{before}}$ *with* $T_R > 0$*. Then*

$$F_R(q_{\text{after}}) \;\leq\; F_R(q_{\text{before}}).$$

(`h_theorem_recognition`)

Proof. By (9), the change in free energy equals $T_R(KL_{\text{after}} - KL_{\text{before}})$. Since $KL_{\text{after}} \leq KL_{\text{before}}$ and $T_R > 0$, we conclude $F_R^{\text{after}} \leq F_R^{\text{before}}$. □

This makes precise the thermodynamic bridge used in the formalization: once a Gibbs/KL decomposition is available and KL contracts under the relevant coarse-graining step, free energy decreases along $\hat{R}$-trajectories. The only new axiom in Lean is the Data Processing Inequality encoding that contraction.

9 Cooling Under an Explicit Energy-Partition Assumption

Definition 9.1 (Implosion process)**.** An *implosion process* is a transition with $J_{\text{after}} \leq J_{\text{before}}$ together with the explicit balance identity

$$J_{\text{before}} - J_{\text{after}} = W + Q, \tag{10}$$

where $W \geq 0$ is work extracted and $Q \geq 0$ is heat released. (`ImplosionProcess`)

Theorem 9.2 (Conditional implosion cooling)**.** *If* $W < J_{\text{before}} - J_{\text{after}}$ *(less work extracted than cost released), then* $Q > 0$*.* (`implosion_cooling_forced`)

Proof. From (10), $Q = (J_{\text{before}} - J_{\text{after}}) - W > 0$. □

The theorem is deliberately modest: it proves a cooling consequence from the balance identity, but it does not itself derive that identity from T3 or any more primitive conservation law. When heat $Q > 0$ leaves the system at recognition temperature $T_R > 0$, the system entropy change satisfies $\Delta S = -Q/T_R < 0$ (`implosion_entropy_decrease`). Under that assumption, implosion can do work *and* cool the working medium simultaneously because the cost decrease exceeds the work extracted.

By contrast, explosion (J increasing) requires energy input and produces heating within the same balance model.

10 The φ-Spiral as a Unique Self-Similar Candidate

Consider logarithmic spirals $r(\theta) = r_0 \cdot g^{\theta/(2\pi)}$ with growth factor $g > 0$. Each full turn scales the radius by g, incurring cost $J(g)$ per turn.

Theorem 10.1 ($J(\varphi)$ from the golden equation)**.** *If $g^2 = g + 1$ and $g > 0$, then $J(g) = g - \frac{3}{2}$.* (`Jcost_of_phi`)

Proof. From $g^2 = g + 1$: $g^{-1} = g - 1$. Then $J(g) = \frac{1}{2}(g + g - 1) - 1 = g - \frac{3}{2}$. □

Theorem 10.2 (Uniqueness of φ)**.** *The golden ratio $\varphi = (1 + \sqrt{5})/2$ is the unique positive solution of $g^2 = g + 1$.* (`phi_unique_self_similar`)

Proof. Factor: $g^2 - g - 1 = (g - \varphi)(g - \psi)$ where $\psi = (1 - \sqrt{5})/2 < 0$. Since $g > 0$, only $g = \varphi$ satisfies the equation. □

Theorem 10.3 (φ-spiral in the self-similar class)**.** *The logarithmic spiral with growth factor φ is the unique spiral satisfying the self-similarity recursion $g^2 = g + 1$ among all $g > 0$. Its cost per turn is $J(\varphi) = \varphi - 3/2 \approx 0.118$.* (`phi_spiral_is_unique_optimal`)

This identifies φ as the sole positive growth factor within the restricted self-similar recursion $g^2 = g + 1$. It therefore provides an RS-motivated candidate geometry for coherent spiral confinement. Global optimality over all spiral families or all three-dimensional confinement geometries is not proved here.

11 Implications

11.1 Fusion

The RS φ-scheduled ignition formalization [2] already shows lower effective ignition temperature via coherence: $T_{\text{eff}} = T/S^2$ where S is the coherence enhancement factor. Theorem 10.3 does not prove that φ-spiral confinement is globally optimal among all geometries, but it does single out a unique self-similar spiral candidate within the present RS framework. The resulting engineering prediction is concrete: spiral confinement geometries should be benchmarked against linear pinch designs for ignition threshold, stability, and input energy.

11.2 Propulsion

The `SpiralField` module defines the φ-spiral wavefield $r(\theta) = r_0 \cdot \varphi^{\kappa\theta/(2\pi)}$. Theorem 5.1 proves inward motion costs strictly less than outward motion. This is a local cost comparison, not a full fluid-dynamical efficiency theorem. Still, if a propulsion architecture can be modeled as repeated ratio updates aligned with the inward cost direction, the RS formalism predicts an advantage for centripetal vortex stages over outward-expulsive ones. Implosion turbines and related vortex machines are therefore natural RS test cases rather than settled winners.

11.3 Thermodynamics

The statement that implosion cools while doing work appears here in conditional form. The RS chain is:

1. $\hat{R}$ decreases J-cost (axiom, centripetal by Theorem 3.1).
2. Under the Gibbs/KL bridge and KL contraction, F_R decreases along $\hat{R}$-trajectories (Theorem 8.2).

3. Under the explicit balance identity (10), cost decrease > work extracted ⇒ heat released (Theorem 9.2).
4. If that heat leaves the system at $T_R > 0$, then $\Delta S < 0$ (cooling).

This does not establish perpetual motion or a complete thermodynamic theory of vortex machinery. It isolates the exact assumptions under which the sentence "implosion cools while doing work" follows inside the present formalism.

12 Formalization Status

Table 1: Lean 4 formalization summary (`RecognitionScience.Foundation.ImplosionDynamics`).

Metric	Value	Note
Lines of Lean code	473	single module
Theorems / lemmas / defs	39	
`sorry` proof terms	0	fully machine-verified
Axioms	1	Data Processing Inequality for Section 8
Build jobs	7812	clean, 0 errors

The single axiom (`data_processing_inequality`) is a standard result in information theory stating that KL divergence cannot increase under Markov maps [3]. It belongs to the same category as the 12 remaining axioms in the RS codebase (external mathematical facts).

The geometric core (Sections 2–7) is unconditional once the RS cost law is accepted. Section 8 adds the `data_processing_inequality` axiom together with explicit free-energy decomposition hypotheses. Section 9 uses an explicit balance identity inside `ImplosionProcess`. Section 10 proves uniqueness only within the self-similar class $g^2 = g + 1$.

13 Discussion

The central result of this paper—gradient centripetality (Theorem 3.1)—is pure mathematics. It makes no empirical assumptions. Given *any* cost function of the form $J(x) = \frac{1}{2}(x + x^{-1}) - 1$, the gradient field is centripetal. Since the RCL uniquely forces this form, the conclusion is inescapable: **cost-minimizing dynamics in Recognition Science is implosion.**

The rest of the manuscript should be read as a structured extension program on top of that core. The present paper does *not* yet prove that all vortex machines outperform combustion engines, nor that the φ spiral is a global optimum among all possible geometries. What it does prove is that any dynamics faithfully modeled as J-cost descent is inward-seeking, and that standard thermodynamic and information-theoretic hypotheses turn this geometry into free-energy decrease and possible cooling-under-work conditions.

That is already enough to generate falsifiable consequences. The next decisive steps are to derive the balance identity from ledger conservation inside Lean, to couple cost descent to explicit fluid and plasma dynamics, and to compare spiral and non-spiral architectures experimentally. If those bridges succeed, implosion-based machinery will move from interpretive analogy to theorem-backed engineering. At present, the mathematical core is established and the experimental program is clear.

Acknowledgments

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References

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